

Haisen Zhao

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Research Interests: Geometric Computing; AI-Driven Design and Manufacturing; Computational Fabrication



Academic Appointments and Education

Oct. 2022-Present	Professor, School of Computer Science and Technology, Shandong University
Sep. 2021-Sep. 2022	Postdoctoral Researcher, Institute of Science and Technology Austria; host: Prof. Bernd Bickel
Mar. 2019-Aug. 2021	Postdoctoral Researcher, Paul G. Allen School of Computer Science & Engineering, University of Washington; host: Prof. Adriana Schulz
Jan.-Feb. 2019	Research Intern, Center on Frontiers of Computing Studies, Peking University; advisor: Prof. Baoquan Chen
2014-2018	Ph.D. in Computer Science and Technology, Shandong University; advisor: Prof. Baoquan Chen
2011-2014	M.S. in Computer Software and Theory, Shandong University
2007-2011	B.Eng. in Digital Media Technology, Shandong University

Research Funding

2025-2028	National Natural Science Foundation of China, No. 62472257 (Principal Investigator)
2024-2027	National Key R&D Program of China, No. 2024YFB3309502 (Principal Investigator)
2024-2027	National Natural Science Foundation of China, No. U23A20312 (Participant)
2023-2025	Science Fund Program for Excellent Young Scientists (Overseas) (Principal Investigator)
2023-2025	National Key R&D Program of China, No. 2022ZD0160805 (Participant)

Publications

* Corresponding author. Publications are listed in reverse chronological order.

2026

Shudan Guo, Fanchao Zhong, Liyuan Wang, **Haisen Zhao***. *Neural Field-Based Sequence Planning for Additive-Subtractive Hybrid Manufacturing*. Eurographics Symposium on Geometry Processing (SGP), Poster Track, 2026.

Wenpeng Xu, Dongxia Wang, Jinglong Zhang, Jiahe Hu, Hao Xu, Liuchao Jin, Haitao Ye, **Haisen Zhao**, Wei-Hsin Liao. *Enhanced Energy Absorption in Re-Entrant Honeycombs via Elliptical Double-Arrow Hybrid Design*. Thin-Walled Structures, 2026.

Pengfei Wang, Qinghao Guo, **Haisen Zhao**, Shiqing Xin, Shuangmin Chen, Changhe Tu, Wenping Wang. *Manifold k-NN: Accelerated k-NN Queries for Manifold Point Clouds*. ACM SIGGRAPH, Journal Track, 2026.

Fanchao Zhong, Yao Zhang, Guanze Xin, Peng-Shuai Wang, Lin Lu, Changhe Tu, **Haisen Zhao***. *DeepMill++: Neural Guidance Meets Rasterization for Efficient Accessibility Analysis*. ACM SIGGRAPH, Conference Track, 2026.

Fei Yu, Yu Liu, Luyang Tang, Mingchao Sun, Zengye Ge, Rui Bu, Yuchao Jin, **Haisen Zhao**, He Sun, Yangyan Li, Mu Xu, Wenzheng Chen, Baoquan Chen. *From Orbit to Ground: Generative City Photogrammetry from Extreme Off-Nadir Satellite Images*. CVPR Findings, 2026.

Yuyun Wei, **Haisen Zhao**, Jia Cui, Meng Qi. *Rich and Cross-Fused Feature Embedding for Few-Shot Point Cloud Semantic Segmentation*. Engineering Applications of Artificial Intelligence, 2026.

Fei Yu, Shudan Guo, Shiqing Xin, Beibei Wang, **Haisen Zhao***, Wenzheng Chen. *Recovering 3D Shapes from Ultra-Fast Motion-Blurred Images*. International Conference on 3D Vision (3DV), 2026.

2025

Liyuan Wang, Fanchao Zhong, Manyi Li, **Haisen Zhao***. *DHMP: A Large 3D Model Dataset for Hybrid Manufacturing Process Planning*. Journal of Computer-Aided Design & Computer Graphics, 2025 (in Chinese).

Fanchao Zhong, Zhenmin Zhang, Liyuan Wang, Xin Yan, Jikai Liu, Lin Lu, **Haisen Zhao***. *Waste-to-Value: Reutilized Material Maximization for Additive and Subtractive Hybrid Remanufacturing*. ACM SIGGRAPH Asia, Journal Track, 2025.

Zhenmin Zhang, Shuai Feng, Hao Xu, Lujiaoyang Fu, Lin Lu, Jianwei Guo, **Haisen Zhao***. *Chapper: Carvable Hull-and-Pack for Subtractive Manufacturing*. ACM SIGGRAPH Asia, Journal Track, 2025.

Chengdong Wei, Shuai Feng, Hao Xu, Qidong Zhang, Songyang Zhang, Zongzhen Li, Changhe Tu, **Haisen Zhao***. *Carvable Packing of Revolved 3D Objects for Subtractive Manufacturing*. Graphical Models, Special Issue on Add-and-Sub Manufacturing, 2025.

Fanchao Zhong, Yang Wang, Peng-Shuai Wang, Lin Lu, **Haisen Zhao***. *DeepMill: Neural Accessibility Learning for Subtractive Manufacturing*. ACM SIGGRAPH, Conference Track, 2025.

Tianyang Xue, Longdu Liu, Lin Lu*, Paul Henderson, Pengbin Tang, Haochen Li, Jikai Liu, **Haisen Zhao**, Hao Peng, Bernd Bickel. *MIND: Microstructure INverse Design with Generative Hybrid Neural Representation*. ACM SIGGRAPH, Conference Track, 2025.

Zheng Zhang, Yifei Li, Kang Wu, **Haisen Zhao**, Xu Liu, Xiang Wang, Ligang Liu, Xiao-Ming Fu. *Carving Shapes with Ruled Surfaces for Rough Machining*. Computers & Graphics, 2025.

Runze Xue, Fei Yu, Baohang Zhou, Jialu Wang, Fan Zhong, Qiong Zeng, **Haisen Zhao**. *WireSculptor: Interactive Guided Bending Workflow for Novice-Friendly Wire Sculpture Fabrication*. International Conference on Extended Reality (ICXR), 2025.

2024

Qibing Wu, Zhihao Zhang, Xin Yan, Fanchao Zhong, Yueze Zhu, Xurong Lu, Runze Xue, Rui Li, Changhe Tu, **Haisen Zhao***. *Tune-It: Optimizing Wire Reconfiguration for Sculpture Manufacturing*. ACM SIGGRAPH Asia, Conference Track, 2024.

Zhenmin Zhang, Zihan Shi, Fanchao Zhong, Kun Zhang, Wenjing Zhang, Jianwei Guo, Changhe Tu, **Haisen Zhao***. *Continuous Toolpath Optimization for Simultaneous Four-Axis Subtractive Manufacturing*. Computer Graphics Forum, 2024.

2023

Fanchao Zhong, **Haisen Zhao***, Haochen Li, Xin Yan, Jikai Liu, Baoquan Chen, Lin Lu*. *VASCO: Volume and Surface Co-Decomposition for Hybrid Manufacturing*. ACM Transactions on Graphics (SIGGRAPH Asia Journal Track), 2023.

Fanchao Zhong, Yonglai Xu, **Haisen Zhao***, Lin Lu*. *As-Continuous-As-Possible Extrusion-Based Fabrication of Surface Models*. ACM Transactions on Graphics (presented at SIGGRAPH), 2023.

2022

Haisen Zhao, Max Willsey, Amy Zhu, Chandrakana Nandi, Zachary Tatlock, Justin Solomon, Adriana Schulz. *Co-Optimization of Design and Fabrication Plans for Carpentry*. ACM Transactions on Graphics (presented at SIGGRAPH), 2022.

Ben Jones, Yuxuan Mei, **Haisen Zhao**, Taylor Gotfrid, Jennifer Mankoff, Adriana Schulz. *Computational Design of Knit Templates*. ACM Transactions on Graphics (presented at SIGGRAPH), 2022.

2021

Molly Carton, Chandrakana Nandi, Adam Anderson, **Haisen Zhao**, Eva Darulova, Dan Grossman, Jeffrey I. Lipton, Adriana Schulz, Zachary Tatlock. *A Roadmap Towards Parallel Printing for Desktop 3D Printers*. Solid Freeform Fabrication Symposium (SFF), 2021.

Haisen Zhao, Yash Talwekar, Wenqing Lan, Chetan Sharma, Daniela Rus, Adriana Schulz, Jeffrey I. Lipton. *Robotic Jigsaw: A Non-Holonomic Cutting Robot and Path Planning Algorithm*. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021.

James Noeckel, **Haisen Zhao**, Brian Curless, Adriana Schulz. *Fabrication-Aware Reverse Engineering for Carpentry*. Computer Graphics Forum (Symposium on Geometry Processing), 2021.

Yu Xing, Yu Zhou, Xin Yan, **Haisen Zhao**, Wenqiang Liu, Jingbo Jiang, Lin Lu. *Shell Thickening for Extrusion-Based Ceramics Printing*. Computers & Graphics (CAD & Graphics), 2021.

2020

Ali Mahdavi-Amiri, Fenggen Yu, **Haisen Zhao***, Adriana Schulz, Hao Zhang. *VDAC: Volume Decompose-and-Carve for Subtractive Manufacturing*. ACM Transactions on Graphics (SIGGRAPH Asia), 2020.

2019

Chenming Wu, **Haisen Zhao**, Chandrakana Nandi, Jeffrey I. Lipton, Zachary Tatlock, Adriana Schulz. *Carpentry Compiler*. ACM Transactions on Graphics (SIGGRAPH Asia), 2019.

Xin Yan, Cong Rao, Lin Lu, Andrei Sharf, **Haisen Zhao**, Baoquan Chen. *Strong 3D Printing by TPMS Injection*. IEEE Transactions on Visualization and Computer Graphics, 2019.

2018

Haisen Zhao, Hao Zhang, Shiqing Xin, Yuanmin Deng, Changhe Tu, Wenping Wang, Daniel Cohen-Or, Baoquan Chen. *DSCarver: Decompose-and-Spiral-Carve for Subtractive Manufacturing*. ACM Transactions on Graphics (SIGGRAPH), 2018.

2016

Haisen Zhao, Fanglin Gu, Qixing Huang, J. A. Garcia Galicia, Yong Chen, Changhe Tu, Bedrich Benes, Hao Zhang, Daniel Cohen-Or, Baoquan Chen. *Connected Fermat Spirals for Layered Fabrication*. ACM Transactions on Graphics (SIGGRAPH), 2016.

Haisen Zhao, Lin Lu, Yuan Wei, Dani Lischinski, Andrei Sharf, Daniel Cohen-Or, Baoquan Chen. *Printed Perforated Lampshades for Continuous Projective Images*. ACM Transactions on Graphics (presented at SIGGRAPH), 2016.

2015

Haisen Zhao, Lin Lu, Zhitao Bo. *Variational Circular Treemaps for Hierarchical Data*. Journal of Software (PacificVis), 2015 (in Chinese).

2014

Lin Lu, Andrei Sharf, **Haisen Zhao**, Yuan Wei, Qingnan Fan, Xuelin Chen, Yann Savoye, Changhe Tu, Daniel Cohen-Or, Baoquan Chen. *Build-to-Last: Strength to Weight 3D Printed Objects*. ACM Transactions on Graphics (SIGGRAPH), 2014.

2012

Haisen Zhao, Chenglei Yang, Lin Lu, Xiaoting Wang, Yijun Yang, Xiangxu Meng. *An Algorithm for Visibility Computation of Points Based on Voronoi Diagrams*. Computer-Aided Design & Computer Graphics (CAD/CG), Best Student Paper, 2012 (in Chinese).

Doctoral Dissertation

Haisen Zhao. *Geometric Research and Applications in Additive and Subtractive Manufacturing*. Ph.D. thesis, Shandong University, 2018 (in Chinese). CCF Outstanding Doctoral Dissertation Award, 2019.

Teaching

2023-2026	Undergraduate Thesis (Design), undergraduate course supervision
2023-2025	Computer Graphics, undergraduate course
2023-2025	Human-Computer Interaction Techniques, undergraduate course
2024-2025	Advanced Human-Computer Interaction Techniques, graduate course
2024	Foundations of Computer Science, undergraduate course

Selected Talks and Academic Activities

Aug. 2026	Invited talk, "Review of Frontier Hot Topics in Intelligent CAM," CAD/CG 2026
Jul. 2026	Invited talk, "Complex Geometric Computation for Subtractive Manufacturing," GDC 2026
Oct. 2021	Invited talks at the University of California, Merced and the Institute of Automation, Chinese Academy of Sciences
Jul. 2021	Invited talk, USTC Summer School, "Advances in Computer Graphics"
Nov. 2019	Conference presentation, ACM SIGGRAPH Asia, Brisbane, Australia
Aug. 2018	Conference presentation, ACM SIGGRAPH, Vancouver, Canada
Jul. 2016	Two conference presentations, ACM SIGGRAPH, Anaheim, USA
Apr. 2015	Conference presentation, IEEE Pacific Visualization, Hangzhou, China

Academic Service

Reviewer: ACM SIGGRAPH and ACM SIGGRAPH Asia

Reviewer: ACM Transactions on Graphics; Computer-Aided Design; Computers & Graphics; The Visual Computer

Reviewer: IEEE Computer Graphics and Applications; IEEE Transactions on Automation Science and Engineering

Reviewer: Eurographics; 3D Printing and Additive Manufacturing; Frontiers of Information Technology & Electronic Engineering

Patents and Software Copyright

Method for planning 3D printing paths based on Fermat spirals. US Patent US10639850B2. Inventors: Baoquan Chen, Daniel Cohen-Or, Hao Zhang, Haisen Zhao.

A printed perforated lampshade generation method for halftone projection. Chinese Patent Application 201410420912.4. Inventors: Lin Lu, Baoquan Chen, Yuan Wei, Haisen Zhao.

An improved printed perforated lampshade generation method for halftone projection. Chinese Patent Application 201610150875.9. Inventors: Lin Lu, Haisen Zhao, Baoquan Chen, Yuan Wei.

A customized 3D printing method for hollowing a model surface. Chinese Patent Application 201510656994.7. Inventors: Lin Lu, Zhitao Bo, Haisen Zhao, Baoquan Chen.

A 3D printing toolpath planning method based on Fermat spirals. Chinese Patent Application 201610242579.1. Inventors: Baoquan Chen, Daniel Cohen-Or, Hao Zhang, Haisen Zhao.

An internal-structure optimization method for 3D printing. Chinese Patent Application 201410230442.5. Inventors: Lin Lu, Changhe Tu, Baoquan Chen, Xuelin Chen, Haisen Zhao, Yuan Wei, Qingnan Fan.

Interactive system and method for a smartphone-based 3D museum. Chinese Patent Application 201210057586.6. Inventors: Li Liu, Chenglei Yang, Haisen Zhao, Bing Sun, Xiangxu Meng.

SDUsoft virtual museum design system based on multi-touch technology V1.0. Software Copyright 2013SR007412.

Honors and Awards

2020	First Prize, Shandong Natural Science Award, Shandong Province
2019	CCF Doctoral Dissertation Award
2018	Innovation Capacity Improvement Scholarship, Shandong University
2017	Excellent Graduate Student Scholarship, Shandong University
2016	President Scholarship, Shandong University
2016	National Scholarship, Ministry of Education of China
2012	Best Student Paper, CAD/CG 2012